

Thesis Structures.

Literature Review Chapter.

Writing up Bachelor Thesis

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Agenda and **learning outcomes:**

1. Thesis Structures

2.1 Literature Review - WHY?

2.2 Literature review - HOW? search

2.3 Literature review - HOW? write up

Thesis Structures

Default thesis structure

Title page

Table of contents

Tables of figures and terms*

Dedication*

(Acknowledgements)*

Abstract (informative) 150-250 words

(Acknowledgements)*

Body:

- Introduction
- Literature Review
- Design & Methodology
- Implementation & Results
- Analysis and Discussion
- Conclusion

(Acknowledgements)*

Bibliography/References

Appendices

40 pages

Alternative contents #1 (system-development thesis)

describe:

- the functionality of the system to be designed;
- the parameters affecting the performance of the system;
- the limitations and restrictions of the system to be designed.

Title page

Table of contents

Tables of figures and terms*

Dedication*

Abstract (informative)

Body:

- Introduction (motivation, literature review, problem statement, assumptions, background)
- Theoretical Part / Benchmarking
- Evaluation / Experimental Study (experimental setup, methodology)
- Conclusion and Discussion

(Acknowledgements)*

Bibliography/References

Appendices

Alternative contents #2 (research-based thesis; Math thesis)

describe:

- the goals of the investigation to be carried out in the project;
- the criteria or metrics to be used to assess the outcome of the investigation.

Title page

Table of contents

Tables of figures and terms*

Dedication*

Abstract (informative)

Body:

- Introduction
- Problem and Discussion
- Methods and Results
- Concluding Discussion

(Acknowledgements)*

Bibliography/References

Appendices

Startup as a thesis

Chapters additional to your main thesis:

- Competitor analysis (direct and indirect, table)
- Market analysis
- Product Market Fit
- Monetization model
- Financial plan, unit economics and scaling

Your thesis

Default

Title page

Table of contents
Tables of figures and terms*
Dedication*

Abstract (informative)

Body:

- Introduction
- Literature Review
- Design & Methodology
- Implementation & Results
- Analysis and Discussion
- Conclusion

(Acknowledgements)*
Bibliography/References
Appendices

System development

Title page

Table of contents
Tables of figures and terms*
Dedication*

Abstract (informative)

Body:

- Introduction (motivation, literature review, problem statement, assumptions, background)
- Theoretical Part / Benchmarking
- Evaluation / Experimental Study (experimental setup, methodology)
- Conclusion and Discussion

(Acknowledgements)*
Bibliography/References
Appendices

Math thesis

Title page

Table of contents
Tables of figures and terms*
Dedication*

Abstract (informative)

Body:

- Introduction
- Problem and Discussion
- Methods and Results
- Concluding Discussion

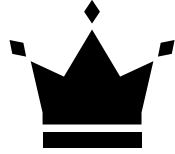
(Acknowledgements)*
Bibliography/References
Appendices

Startup

Additional:

- Competitor analysis
- Market analysis
- Product Market Fit
- Business model
- Financial plan and scalability

YOU DECIDE



HOW:

Consider:

1. IU requirements
2. Domain-specific structures
3. Structures of your supervisors' recent papers.

Literature Review - WHY?

Literature Review - WHY?

Reason 1

Once upon a time

There has long been this problem (or recently emerged).

This is why it is important.

Previously it has been addressed in such and such ways, which had their benefits, but also limitations.

or

No one has ever addressed this problem.

or

No one has ever approached this problem with this method.

or

No one has ever used this method on this population.

So, we decided to give it a try. Particularly, we addressed these RQs (or tested these Hypotheses): ...

To answer those research questions/to test those hypotheses , this is what we did.

And this is what we found.

This is how our results are better or worse than previous ones. And this is why.

To conclude, our point is that ...



... graduated and lived happily ever after.

1. RQ or Hypothesis

A: Is Spring Framework more efficient in enterprise application development than Java EE?

B: Spring Framework is more efficient in enterprise application development than Java EE.

To address RQ or H:

2. Technology ?

Hardware: PC

Software: IDE, jdk, and free space for libraries

3. Which data?

Cons and pros of both Spring and Java EE

4. How to collect them?



5. How to analyze them?



Your thesis

- 1. What is your RQ or Hypothesis?**
- 2. Which technology will be used to answer your RQs or to test your Hypotheses?**
- 3. Which data to collect to answer your RQs or to test your Hypotheses?**
- 4. How to collect those data?**
- 5. How to analyze them?**

Literature Review - WHY?

Reason 2



«How do your results compare with previous results?»

«What does literature say about it?»

«Which methods were you choosing from?»

«What other methods do you know?»

«Why did you choose this method?»

«Why these metrics?»

«What does this metric mean?»



Literature review - HOW literature search

2 ways

Systematic Literature Review

PRISMA protocol

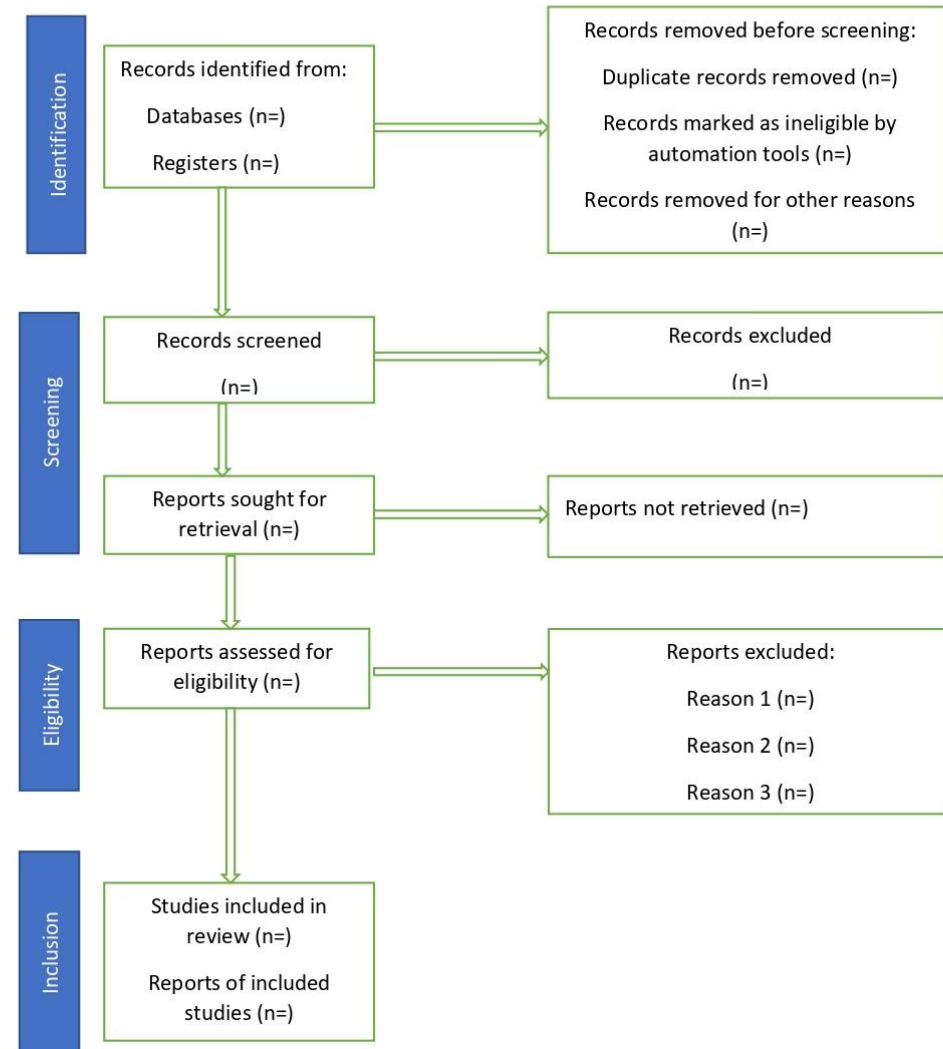
1. Databases (IEEE Xplore /Scopus /ScienceDirect)
2. Replicable search results

Steps:

1. Define RQs for the literature review.
2. Define key words
3. Design search strings
4. Perform search in databases
5. Refine strings
6. Download and store search results
7. Define inclusion/exclusion criteria
8. Screen titles and abstract
9. Scan papers for quality
10. Extract answers to your RQs

Weidt, F., & Silva, R. (2016). Systematic literature review in computer science -a practical guide. *Relatórios Técnicos Do DCC/UFJF*, 1(8), 1-7.

Here is the PRISMA 2020 flow diagram



Literature Review

Identity research in mathematics education

[L Darragh](#) - Educational studies in mathematics, 2016 - Springer

... **identity** but not about either **mathematics identity** or the professional **identities** of **mathematic** teachers... These other "types" of **identity** included gender-**identity**, racial- or ethnic-**identity**, ...

★ Сохранить Цитировать Цитируется: 454 Похожие статьи Все версии статьи (10)

Where to get the first paper?

A: Your supervisor

Forward snowball:

B: Random search:

1. key words
2. a database or an AI

Backwards snowball

References

- Anderson, R. (2007). Being a mathematics learner: Four faces of identity. *Mathematics 1* 7–14.
- Andersson, A. (2011). A 'Curling teacher' in mathematics education: Teacher identities and pe
ment. *Mathematics Education Research Journal*, 23(4), 437–454. doi:10.1007/s13394-011
- Bartholomew, H., Darragh, L., Ell, F., & Saunders, J. (2011). 'I'm a natural and I do it for l
students' accounts of studying mathematics. *International Journal of Mathematical Educ
and Technology*, 42(7), 915–924. doi:10.1080/0020739x.2011.608863
- Bernstein, B. (2000). *Pedagogy, symbolic control and identity: Theory, research, critique*. Lan
Littlefield.
- Bishop, J. (2012). "She's always been the smart one. I've always been the dumb one":
mathematics classroom. *Journal for Research in Mathematics Education*, 43(1), 34–74.

Cultural identity

hide

Article [Talk](#)

From Wikipedia, the free encyclopedia

1

ena

entity and
experience

sitions



This article's **tone or style may not reflect**
See Wikipedia's [guide to writing better](#)
when to remove this message)

Cultural identity is a part of a person's [identity](#), or their self-con-
[perception](#), and is related to [nationality](#), [ethnicity](#), [religion](#), [social](#)
[locality](#), [gender](#), or any kind of social group that has its own disti
cultural identity is both characteristic of the [individual](#) but also of

1. Credible or not? - NO

2. Should be in the Reference list? - NO

Literature review - HOW writing up: summary vs synthesis

A In several previous studies, antagonistic TSA configurations were investigated in various robotic applications such as variable-stiffness robotic joints as explored by Shisheie *et al.* [1], Popov *et al.* [7], G Palli *et al.* [8], and Park *et al.* [9]. However, using two motors for single joint actuation is not cost-effective, and the non-linear nature of TSA contraction with twisting, complicates control over antagonistic TSA-based systems. If not actuated properly, these systems may suffer from increased string wear due to in-fighting between them, as well as from cable slack and loss of tracking accuracy as the result of it.

Source: M. Khan, B. Suthar, I. Gaponov, J. H. S. Ryu. "Single-motor-based bidirectional twisted string actuation with variable radius pulleys." *IEEE Robotics and Automation Letters* 4.4 (2019): 3735-3741. [Link](#)

B In light of results presented by [31], Aljabri *et al.* [32] conducted study having a similar research objective to [31] but the sorting algorithms were generic merge sort and classically optimized 3-way partitioning quick sort. The algorithms used by [32] were implemented in C++ and tested on desktop CPU. The researchers used 31 different datasets of randomly generated integer numbers. The main finding of this study was that merge sort is better against a highly optimized quick sort in terms of power consumption.

Source: G. Dlamini, F. Jolha, Z. Kholmatova, G. Succi. "Meta-analytical comparison of energy consumed by two sorting algorithms." *Information Sciences* 582 (2022): 767-777. <https://www.sciencedirect.com/science/article/pii/S0020025521009981>

Literature review - HOW writing up: structures

SLR

Thesis RQ:

How can recommender systems apply to dashboards?

Preamble

Thesis problem and its importance (a way apply of recommender systems to dashboards). Purpose of the chapter; overview of the structure of the chapter.

Literature search
and selection
(PRISMA)

1. Literature review methodology

- RQs for LR
- Search strings and databases
- Papers selection process; inclusion and exclusion criteria
- Papers selection results

Literature analysis

2. Papers review

- On recommender systems
 - Content-based methods
 - Collaborative filtering
 - Demographics methods
 - Knowledge-based methods
 - Community-based methods
 - Hybrid methods
- On dashboard types
 - nnn
 - mmm

Relevance

Conclusion of the chapter – which recommender systems are relevant to my project and why.

Literature review

Preamble

Literature analysis

Relevance

RQ/Hypothesis/Problem:

How can the **Image-Based Virtual Try-On approach** be improved in terms of **characteristic preserving and clothing types variety**?

Outline:

Chapter 2.

This chapter presents an overview of **previous research, methods comparison**. *Describe literature search process: RQs, databases, key words, search strategies, search results.

Section 2.1 compares two currently used methods of **Section 2.2 elaborates on the Performance metrics**. Section 2.3???

Section 2.1. Overview of the approaches and methods.

There are two methods related to: **name of the method**, **name of the method**. both target virtual ... but use different procedures.

Section 2.1.1. 3D-based Virtual Try-On

Definition. What is the approach in a nutshell. [reference]. [reference] claim that
. Besides...[reference], [reference], [reference] found that However, limitations [reference], [reference]. Conclusion re applicability of the methods.

Section 2. 1.2. Image Based Virtual Try-On

Definition. What is the method in a nutshell. How it is different from the previous approach. Relate methods:

Sub section clothes ()

Sub-section human wearing clothes ()

Sub-section ???

Conclusion re applicability, including for this project.

Section 2.1 conclusion: two methods exist. In this study we relied on method A because

Section 2. Metrics overview?

Section conclusion.

Chapter conclusion – what does it mean to my work, which **methods** and **metrics** are relevant to my project and why.

Useful links:

Previous years theses: <https://e.lanbook.com/vkrs>

SLR:

1. <https://www.prisma-statement.org>
2. https://www.researchgate.net/publication/302924724_Guidelines_for_performing_Systematic_Literature_Reviews_in_Software_Engineering
3. https://www.researchgate.net/publication/320704338_Systematic_Literature_Review_in_Computer_Science_-_A_Practical_Guide

IEEE references guide:

<https://docs.google.com/document/d/1j1L96U2NagwWI9MEVDNVKt9pXxRzTH7h3krI3Mb6wZE/edit?tab=t.0>

Book a writing consultation:

<https://planerka.app/oksana-zhirosh-yxrbr4>